This means that we have built one of our two vectors. The second follows as a linear combination of the others:

$$\tilde{\boldsymbol{F}} = \boldsymbol{b}_0 - \xi\tilde{\boldsymbol{G}} - (g - \gamma\xi)\sigma^{-1}\boldsymbol{b}_1.$$

□

The fact that the Hörmander condition holds is, as we know from Chap. 6, only part of the story. However based on known results for Langevin dynamics, we conjecture that ergodicity will hold if (i) the Hörmander condition holds, (ii) U is sufficiently smooth, and (iii) the configurational phase space is compact, e.g. periodic boundary conditions are employed. However we stress that each possible method will ultimately need to be carefully and systematically checked to verify the ergodic properties.

8.4.1 *Nosé-Hoover-Langevin: An Ergodic Weakly Coupled Thermostat*

The Nosé-Hoover-Langevin (NHL) method is based on a simple idea: replace the "chain" in the Nosé-Hoover Chain, whose sole purpose is to maintain a Gaussian distribution in the auxiliary variable, by a stochastic Langevin-type thermostat. The method was first proposed in [323]. The proof of ergodicity (more precisely the confirmation of the Hörmander condition), for a problem with harmonic internal interactions, was given in [226] and we roughly follow the treatment from this paper.

The Nosé-Hoover-Langevin equations are

$$\mathrm{d}\boldsymbol{q} = \boldsymbol{M}^{-1}\boldsymbol{p}\mathrm{d}t, \tag{8.27}$$

$$\mathrm{d}\boldsymbol{p} = -\nabla U(\boldsymbol{q})\mathrm{d}t - \xi\boldsymbol{p}\mathrm{d}t, \tag{8.28}$$

$$\mathrm{d}\xi = \mu^{-1}[\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p} - \mathrm{N_d k_B T}]\mathrm{d}t - \gamma\xi\mathrm{d}t + \sqrt{2\mathrm{k_B T}\gamma\mu^{-1}}\mathrm{dW}, \tag{8.29}$$

where $(\boldsymbol{q},\boldsymbol{p}) \in \mathcal{D}$ with $\xi \in \mathbb{R}$. Let us emphasize that the principal features of this thermostat: (1) it is essentially Nosé-Hoover plus a perturbation, (2) it incorporates a single Wiener process, (3) it has two free parameters γ and μ.

Let us first demonstrate that the Nosé-Hoover-Langevin method, is, in fact, a thermostat. We observe that we have already shown that the Nosé-Hoover method preserves the extended canonical distribution $\rho_\beta \exp(-\beta\mu\xi^2/2)$. Therefore, by the additivity property, all that is needed is to show that the additional part

$$\mathrm{d}\xi = -\gamma\xi\mathrm{d}t + \sqrt{2\mathrm{k_B T}\gamma\mu^{-1}}\mathrm{dW}$$

preserves $\hat{\rho}(\xi) = \exp(-\beta\mu\xi^2/2)$. However, this is obvious from our earlier investigations of Ornstein-Uhlenbeck equations.

In order to show that this thermostat is ergodic, we need to demonstrate that the assumptions presented in Sect. 6.4.4 are valid. Assumption 1(ii) requires us to verify that a Hörmander condition (see Definition 6.1) holds for the Nosé-Hoover-Langevin system. Because of the complexity of the high order commutators, we will work here with the assumption of a quadratic potential in the physical model. In some sense this is the most difficult case for a thermostat, but paradoxically, the assumption facilitates the mathematical analysis. We will assume throughout the following that $\mathrm{N_d} = \mathrm{N_c}$, i.e. that there are no constraints in the system. Our potential will therefore be assumed to be

$$U(\boldsymbol{q}) = \frac{1}{2}\boldsymbol{q}^T\boldsymbol{A}\boldsymbol{q},$$

for $\boldsymbol{A}$ a symmetric, positive definite $\mathrm{N_c} \times \mathrm{N_c}$ matrix, with

$$\nabla U(\boldsymbol{q}) = \boldsymbol{A}\boldsymbol{q}.$$

We will further assume $\boldsymbol{M} = \boldsymbol{I}$. If this is not the case, then one may introduce the rescaling $\tilde{\boldsymbol{q}} = \boldsymbol{M}^{-1/2}\boldsymbol{q}, \tilde{\boldsymbol{p}} = \boldsymbol{M}^{1/2}\boldsymbol{p}$ in order to rewrite the system in a form that does not involve $\boldsymbol{M}$. Thus the system under study is

$$\mathrm{d}\boldsymbol{q} = \boldsymbol{p}\mathrm{d}t, \tag{8.30}$$

$$\mathrm{d}\boldsymbol{p} = -\boldsymbol{A}\boldsymbol{q}\mathrm{d}t - \xi\boldsymbol{p}\mathrm{d}t, \tag{8.31}$$

$$\mathrm{d}\xi = \mu^{-1}[\|\boldsymbol{p}\|^2 - \mathrm{N_d}\mathrm{k_B}\mathrm{T}]\mathrm{d}t - \gamma\xi\mathrm{d}t + \sigma\mathrm{d}\mathrm{W}. \tag{8.32}$$

Let $\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_{\mathrm{N_c}}$ be the eigenvectors of $\boldsymbol{A}$ with $\lambda_1, \ldots, \lambda_{\mathrm{N_c}}$ the corresponding eigenvalues. Observe that if $\boldsymbol{q}_0 \cdot \boldsymbol{u}_k = \boldsymbol{p}_0 \cdot \boldsymbol{u}_k = 0$, then, using (8.30)–(8.31), we have at this point

$$\boldsymbol{u}_k \cdot \boldsymbol{p} = 0, \qquad \boldsymbol{u}_k \cdot [-\boldsymbol{A}\boldsymbol{q} - \xi\boldsymbol{p}] = -\lambda_k \boldsymbol{u}_k \cdot \boldsymbol{q} = 0,$$

which implies that $V_k = \{(\boldsymbol{q}, \boldsymbol{p}) | \boldsymbol{u}_k \cdot \boldsymbol{q} = 0, \boldsymbol{u}_k \cdot \boldsymbol{p} = 0\}$ is an invariant submanifold. Therefore we must require that $\boldsymbol{u}_k \cdot \boldsymbol{q}$ and $\boldsymbol{u}_k \cdot \boldsymbol{p}$ do not simultaneously vanish, that is, $(\boldsymbol{q}, \boldsymbol{p}) \in \mathcal{D} = \mathbb{R}^{2\mathrm{N_c}} \setminus \bigcup_{k=1}^{\mathrm{N_c}} V_k$.

Theorem 8.1 *If A is symmetric positive definite and has distinct eigenvalues, then (8.30)–(8.32) satisfies the Hörmander condition on $\mathcal{D} \times \mathbb{R}$.*